

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Structural Works Specifications

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Revision History

Rev No.	Date	Description of Revision
R01	31.05.22	First Issue for Client's Review
R02		
A01		

Review Code & Status (to be filled by Client)		
☐ Code 1	Approved – Further work may proceed.	
☐ Code 2	Approved with comments. – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ Code 3	Reviewed – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ Code 4	Retained for information only	
☐ Code 5	Review Not Required – Do not Resubmit	
Date:	Position:	Sign:

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- MPP-SPC-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
AASHTO	American Association of state Highway Transportation Officials
AGEF	Above Ground End Facilities
ALARP	As Low as Reasonably Practicable
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BOD	Basis Of Design
CP	Cathodic Protection
CPT	Cone Penetrometer Test
CTMS	Custody Transfer Metering Station
DB&B	Double Block and Bleed
DCQ	Daily Contractual Quantity
DEP	Design and Engineering Practice
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
FEED	Front End Engineering Design
FGS	Fire, Gas and Smoke detection systems
GPP	Gas Processing Plant
GTG	Gas Turbine Generators
HAZID	Hazard Identification
HAZOP	Hazard Operability
HEMP	Hazards and Effects Management Process
HFE	Human Factors Engineering
HPU	Hydraulic Power Unit
HRA	Health Risk Assessment
HSES	Health, Safety, Environment and Security
HSSE & SP	Health, Safety, Security, Environment and Social Performance
IEC	International Electro-technical Commission

ISPM	Industry Standard Process Modules
JV	Joint-Venture
LCD	Liquid Crystal Display
LOI	Local Operator Interface
LRL	Lower Range Limit
LRV	Lower Range Value
LIRA	Logistics and Infrastructure Resource Assessment
MAC	Main Automation Contractor
NAG	Non-Associated Gas
OPC	OLE for Process Control
OPPS	Overpressure Protection System
PCD	Process Control Domain
PDMS	Plant Design and Management System
PEPF	Package Engineering, Procurement, and Fabrication
PIG	Pipeline Inspection Gauge
PVC	Polyvinyl Chloride
RTR	Reliability Test Run
SCE	Safety critical elements
SI	International System Units
SIL	Safety Integrity Level
SRB	Sulphate Reducing Bacteria
SIS	Safety Instrumented System
SWB	Steel Wire Braiding
URV	Upper Range Value
USC	US Customary Units
XLPE	Cross Linked Polyethylene
TEG	Tri-ethylene Glycol
TQ	Top Quartile

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

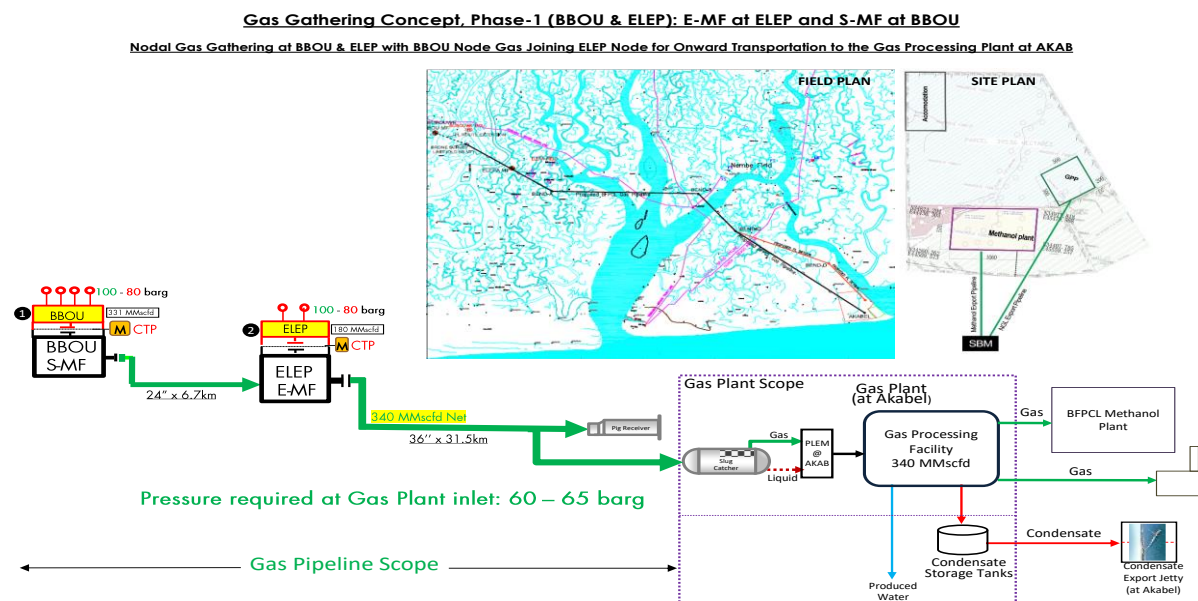


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This Specification covers the requirements for fabrication of structural steel designated to be used for the Project.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

All work shall comply with the latest edition of AISC Code of Standard practice for Steel Buildings and Bridges, AISC Specification for Design Fabrication and Erection of Structural Steel for Buildings, and AWS Structural Welding Code AWS D1.1, and the requirements herein. Where other specifications, standards, or codes are referred to herein, the latest editions are intended.

Employer's requirements shall take precedence over all other requirements.

The following documents are referenced herein and are considered a part of this specification. The latest edition upon date of issuance of the contract shall apply.

- American Institute of Steel Construction (AISC)
- AISC manual of Steel Construction, Allowable Stress Design, Ninth Edition. The manual also contains the following:
 - Specification for Structural Steel Buildings – Allowable Stress Design and Plastic Design.
 - Code of Standard Practice for Steel Buildings and Bridges.
 - Specification for structural Joints using ASTM A325 or A490 Bolts - Approved by the Research Council on Structural Connection of the Engineering Foundation.
 - Specification for Allowable Stress Design of Single Angle Members.
- AISC Manual of Steel Construction, Volume II, Connections ASD/LRFD
- American Society for Testing and Materials (ASTM):

Document Number	Document Title
ASTM A6	Specification for General Requirements for Rolled Steel Plates, Shapes, Set Pilings and Bars for Structural Use.
ASTM A-36	Plate and Structural Steel Shapes
ASTM A-53	Specification for Pipe, Steel, Black and Hot Dipped Zinc-Coated Welded and Seamless Steel Pipe
ASTM A-123	Specification for Zinc (Hot-Pipe Galvanized) Coating on Iron and Steel Products.
ASTM A-153	Specification for Zinc Coating (Hot Dip) on Iron Steel
ASTM A283	Specification for Low and Intermediate Tensile Strength Carbon Steel Plates,

	Shapes and Bars.
ASTM A-385	Practice for Providing High-Quality Zinc Coatings (Hot Dip)
ASTM A-307	Specification for Carbon Steel Bolts and Studs 60, psi tensile Strength.
ASTM A-325	Specification for High Strength Bolts for structural Steel Joints.
ASTM A-501	Specification for Hot-Formed and Welded and Seamless Carbon Steel Standard Tubing
ASTM A-563	Specification for Carbon and Alloy Steel Nuts
ASTM A-569	Specification for Steel, Carbon (0.15 maximum Percent), Hot Rolled Sheet and Strip Commercial Quality
ASTM A-570	Specification for Hot-Rolled Carbon Steel Sheet and Strip Structural Quality.
ASTM A-572	Specification for High Strength, Low Alloy Columbian vanadium Steel of Structural Quality.
ASTM A-611	Specification for Steel Cold-Rolled Sheet, Carbon, Structural
ASTM B-695	Specification for Coating of Zinc Mechanically Deposited on Iron and Steel
ASTM F-436	Specification for Hardened Steel Washers
ASTM F-959	Specification for Compressible-Washer-Type Direct Tension Indicator for use with Structural Fasteners.
ANSI (ASME) B18.23.1	Standard for Bevelled Washers
ANSI 18.22.1	Standard for Plain Washer
ANSI/NAAMM A202.1	Metal Bar Grating Manual

3. Structural Materials

3.1. Plates, Shapes and Fasteners

Unless otherwise specified in writing, all materials shall be of new stock and shall conform to the following Specifications:

Plates and Structural Steel Shapes	ASTM A-36, A572 Grade 50, ASTM A6
Sheets and Strips	ASTM A-570, Grade 33 or 36 or A-611, Grade C
Pipes	ASTM A-501 OR ASTM A-53, Grade B, Type E or S
Low-Carbon Steel	ASTM A-307
High Strength Bolts	ASTM A-325
Welding Electrodes	As identified in AWS D1.1 as an acceptable filler metal for a given steel specification except for Flux Cored Arc welding (FCAW) process. Consumables for designations: E6XT-1, E6XT-5, E6XT-6, E6XT-8, E7XT-1, E7XT-5, E7XT-6, E7XT-8. Other classification is acceptable only with prior approval from the Employer and requires that weld metal composition and Charpy V-Notch toughness be submitted for approval.
Heavy Hex Nuts	ASTM A-563
Hardened Washers	ASTM F-436
DTI Washers	ASTM F-959
Welded Steel Grating	ASTM A-569 (AISI 1020)
Plain Washers	ANSI B 18.22.1, Type A or B
Bevelled Washers	ANSI/ASTME B 18.23.1

3.2. Grating materials

All bar grating shall be welded and serrated with semi-circular serrations formed by punching, as following:

	Minimum (mm)	Maximum (mm)
Radius of Punch	3	5.5
Depth of Cut-out	2	4
Punch Centre-to-Centre	11	12.5

Grating shall be manufactured to National Association of Architectural Metal Manufacturers'

tolerances and specifications listed in the Metal bar Grating Manual ANSI/NAAMM A202.1, except all cross bars be flush or lower than the bearing bars. Unless otherwise specified, the grating shall conform to the standard marking system W-1 9-4-STEEL. Grating shall be galvanized unless other finish specified by the Engineer.

All expanded metal grating shall be fabricated such that the walking surface shall provide skid resistance in all directions.

Grating shall be pre-galvanized before punching unless the Employer specifies other finishing.

Bar and expanded metal grating stair treads shall be fabricated according to the above specification. Each tread and edge of the platform at the top of the stair shall be furnished with a non-skid, cast abrasive nosing. Nosing shall be installed after painting or galvanizing.

No substitutions shall be made in material or size or weight of members unless authorized by the Engineer in writing.

3.3. Welding Consumables

The Shielded Metal-Arc (SMAW), Gas metal arc (GMAW) or Flux Cored Arc (FCAW) welding processes shall perform all welding.

Welding and welding procedures shall be in accordance with AWS D1.1. The written weld procedures and any code required qualification test records should be submitted to the Engineer for approval. Procedure other than those pre-qualified shall be qualified by test as specified in Section 5 of AWS D1.1

Welding shall be performed only by welders, operators and tackers who have been qualified in accordance with Section 5 of WAS D1.1. The welder qualification test records shall be made available to the Employer.

Wind velocity shall not exceed 5 miles per hour in the weld area for gas-shielded processes and 20 miles per hour for self-shielded electrodes.

The interrupted-arc GMAW process shall not be used for anything other than floorplates.

Carbon content (product analysis) shall be limited to 0.35% for forging intended for welding.

Weld-through primers shall not be used unless approved in writing by the Engineer.

4. FABRICATION

Connections shall be as indicated on project drawings and as specified herein. Contractor design connection shall be in general in accordance with the following:

- Project design drawings may indicate beam loading upon which Contractor should base beam connection design. However, unless project design drawings specifically designate design beam loading, the connection for beams shall be designed for a reaction equal to $W/2$ or V , whichever is smaller, where:

W = Allowable uniform load, kips, for the appropriate beam span, as shown in tables for Allowable uniform Loads on Laterally Supported Beams, part 2, of AISC manual of Steel Construction (ASD)

V = Maximum allowable web shear, kips

Bolted beam connections shall consist of at least 2 bolts vertical.

One-sided beam connection or shear-tabs shall not be permitted unless specified on Project design drawings.

Short beams in medium to heavy size shapes may not require connections designed to the above $W/2$ or V criteria. It is preferable to show beam loading on drawings where these criteria may produce unreasonably heavy connections.

- For beam connections (axial loading), Project design drawings shall normally indicate the axial load value which Contractor should use as the basis of connection design. Where design drawings do not indicate such loading values, bolted brace connections shall consist of at least two bolts.

WT and single-angle braces shall normally have single shear connections.

Connections for W-shape braces shall be double shear (plates connected to web)

- For moment connections, primary beam to column (strong axis) moment connections shall be designed to develop the full bending capacity of the weakest connected member, unless the design moment is noted on design drawing. If horizontal and vertical reactions are significant, they shall be shown on Project design drawing. Contractor shall specify the moment connections for a beam to column weak axis on the drawing.
- The gusset plates at the intersection of a column beam and brace shall be connected to both

beam and column. The working point at connection to a column flange shall be the face of column flange and top or bottom of beam flange, depending on whether brace is above or below the beam. The working point at a connection to a column web shall be the centreline of the column web and the top or bottom of the beam flange. The working point of brace to column only (no beam) shall be at the intersection of the face of the column flange or centreline of the web per above.

The working point for K-braces to beams up to 250mm deep shall be the intersection of beam centreline and centreline of bay.

The working point for K-braces to beams 300mm deep shall be the horizontal line either 125mm above beam bottom top and centreline of bay.

- Shop connection may be welded or equivalently bolted. If high strength bolting is used, "turn-of-nut" "calibrated wrench" or "load indicating washer" methods may be used for bolt tightening. Temporary shop erection bolts must be approved and replaced with high-strength bolts. The calibration shall be performed using an approved device capable of indicating actual bolt tension to assure that the minimum required bolt tensions are being obtained in accordance with the AISC Specification for structural Joints using ASTM A-325 Bolts.
- Where galvanized construction is required, ASTM A-325 bolts and DTI washers shall be mechanically galvanized in accordance with ASTM B-695, Class 5D.
- For framed structures, all field connections for columns, beams and frame bracing shall be made with high strength bolts conforming to ASTM A-325 friction type, unless otherwise shown or noted. For floor framing, miscellaneous steel, and where called for on the design drawings, standards bolts conforming to ASTM A-307 shall be used. Included for this use are connections for girts, purlins, ladders, stairs, handrailing, and toe plates.
- Bolt lengths shall conform to criteria outlined in AISC Specification for Structural Joints using ASTM A-325 or A-490 Bolts, Commentary section C2, table 2.
- Bolt holes shall be shop-punched or drilled.
- Shop assembly shall be as complete as possible in order to minimize field connections, and consistent with the feasibility and economy of handling and shipping the assembled units.
- All angles and clips shall be bolted or welded in place and all holes shall be punched for bolted field erection or for fit-up prior to welding for welded field erection.
- The Contractor shall furnish field erection clips and bolts (plus 10% extra) required for field erection.
- In the case of ladders and platforms for columns or other vessels, the purchase order or drawing will indicate whether the platform and ladder supporting clips are to be supplied by the vessel supplier or by the Contractor. If the latter is specified, these clips shall be fabricated and bolted to the proper members before shipment.

- Platform plating shall be fastened to the framing members in the following way:
 - Plates shall be welded to framing members with 5mm fillet welds 37mm long or with 15mm diameter plug weld at intervals not exceeding 450mm. If galvanized construction, floor plate welding should be limited to preassembled units welded in the shop prior to be galvanised.
 - Where plates are shown removable on the drawings, the plates shall be fastened with 10mm diameter galvanized steel countersunk flathead bolt at intervals not exceeding 450mm.
 - If seal welding is specified, it should be made with a 3mm continuous fillet weld around the perimeter of the plate.
- Platform grating shall be fastened by 5mm fillet welds 37mm long at interval not exceeding 450mm at edge support, or with a minimum of welds per panel. Grating shall be tack welded at 450mm centres over intermediate members.
- Removable grating panels shall be fastened to support members by "U" shape clips.
- A minimum of one 12.5mm diameter drain hole shall be provided near the center of each panel of platform floor plate.
- Toe boards shall extend all around the platform, including the edge of adjacent to columns and vessels, and shall be omitted only at passages and access to top of ladders and top of stairs. The gap between toe board and floorplate, both horizontally and vertically, shall not exceed 6mm.
- Handrails shall be assembled and attached to platforms if it is possible to make shipment of the assembled section. All straight handrails shall be shop-assembled in section, with provision for field joints between sections.
- Ladders and hoop guards shall be assembled and shipped as complete units ready to be bolted to the vessels or platforms in the field. Any ladders that cannot be shipped in one piece because of excessive length shall be shop-assembled in sections, with provision for field joints between sections.
- Exposed corners, edges, burrs, weld splatter, or rough spots on handrails, guards, etc., which might cause injury to personnel using the assembled structure, shall be ground or filed smooth.

5. SURFACE PREPARATION AND COATING

Unless otherwise specified, all steel work shall be given a shop prime coat of paint in accordance with the followings, except for steel work to be galvanized, for contact surfaces at high strength bolted connection, for machine finished surfaces, and for surfaces adjacent to field welds.

Prior to coating, all material surfaces except those stated above shall be prepared by blast cleaning in accordance with the Painting Specification no. A942-PPMC-AC-ME-SPC-2203

On the same day the surface is sandblasted, a primer coating in accordance with the Painting Specification no. A942-PPMC-AC-ME-SPC-2203, shall be applied. Minimum dry film thickness shall be in accordance with the above reference specification, except the minimum dry film thickness on surface inaccessible for painting after assembly shall be 4 mils minimum. This includes surface under concrete fireproofing.

Material specified to be galvanized shall be hot dip galvanized after fabrication on the same day the surface is sandblasted or pickled. Galvanizing shall occur before assembly by bolting. Galvanizing procedures shall be in accordance with ASTM A-123, A-153 and A-385.

Machine – finished surface shall be protected against corrosion by rust inhibiting coating which can be easily removed prior to erection or which has characteristics that make removal unnecessary prior erection.

If high strength bolting is used, contact surfaces of members to be joined by friction-type connections must be left unpainted.

6. MARKING FOR ERECTION AND ASSEMBLY FOR SHIPPING

All fabricated steel sections shall be match-marked for field assembly with designating numbers or letters corresponding to the field erection drawings.

A separate number system shall be assigned to each individual complete structure; for example, each building, each equipment support structure, complete pipeway support system all platforms on any large single piece or grouping of similar equipment, etc.

All tagging of bundles of bagged items shall be with cloth tags and waterproofing, or with tags of pressed metal.

Marking of large items shall be welded or stamped to be legible when received at the site or painted on the item with an indelible "paint stick" marker of a sharp contrasting colour to that of the primer.

7. INSPECTION

The Engineer reserves the right to inspect all materials and workmanship covered in this Specification. However, such inspection shall not relieve the Contractor of his responsibility to furnish materials and workmanship in accordance with the Contract requirements. If inspection indicates a weld or part of the material is defective, the Contractor shall remove and replace it at the Contractor's expense.

Shop inspection shall include, but not be limited to, the following:

- Verification of conformance of materials with this Specification and the drawings. The limits of acceptability and repair of surface imperfections for structural steel shall be in accordance with ASTM A-6.
- Visual and dimensional inspection of shop-fabricated structural steel members and assemblies shall be in conformance with this Specification and the Drawings.
- Inspection of high strength bolted connections shall be in accordance with AISC Specification for Structural Joints Using ASTM A-325 or A-490 Bolts.
- Verification of welding procedures, welding operations and welder and taker certificates of qualification shall be in accordance with this Specification and AWS D1.1.
- Inspection of shop-welded shear 4 connection studs shall be in accordance with AWS D1.1, Section 4, Par F.
- Inspection of surface preparation and coating to verify that it conforms to this Specification.
- Non-destructive testing, if required, shall be in accordance with the followings:
 - Visual Inspection of welds in accordance with AWS D1.1.
 - Magnetic particle inspection in accordance with AWS D1.1, Section 6.7.5.
 - Radiographic inspection in accordance with AWS D1.1, Part B, Section 6.
 - Ultrasonic inspection in accordance with AWS D1.1, part C, Section 6.
 - Dye penetrant inspection in accordance with AWS D1.1, Section 6.7.6
- Unless otherwise noted on purchase order, the Contractor shall notify the Engineer at least five days prior to shipping so that arrangement can be made for inspection.

8. SHIPPING AND DELIVERY

It shall be the Contractor's responsibility to provide adequate packing, crating, blocking and/or bracing to prevent damage to the fabricated material while loading.

When galvanized material is specified, the delivery of fabricated steel and grating to the galvanizer after fabrication and to the construction site after galvanizer shall be the responsibility of the Contractor.

Shipping Papers: Each load of fabricated steel delivered to the jobsite (including shipments from the Galvanizer) must be accompanied by a shipping list, identifying each piece on that shipment by piece mark numbers, along with the ship order numbers (or area number on multiple unit project) and the weight of each piece. Fabricated steel from than one shop order shall not be shipped together without prior approval from the Engineer.

Shipments arriving at jobsite without proper shipping list and mark number on the fabricated pieces may be unloaded if requested by the Contractor, but will not be received until the Contractor has visited the site and properly identified all the components.

All required field erection fasteners for a shop order should be delivered prior to, or with the first delivery of shop order, and should be accompanied with a packing list giving description, size, quantities, and proper identification as where they are to be used.

Except as directed by the Employer, stair stringers, trends, grating clips and handrail shall be shipped with related framing system.

9. GUARANTEE

If error in detailing and fabricating work performed by the Contractor prevent accurate field fit-up, the Contractor shall be notified of such defects in writing and given reasonable opportunity to inspect and correct such defective designs, materials, equipment or workmanship at the Contractor's sole expense. If the Contractor fails to make such inspection and correction in a timely fashion, after written notification, the Engineer may then order to the Contractor to perform all the necessary corrections, at Contractor's expenses.